SEMESTER 5 · IMPLEMENTATION STUDY · TMH OSTEONECROSIS AI PROJECT

Segmenting the jaw on planning CT: how we would reproduce IMPACT

A study of the MD Anderson IMPACT paper (medRxiv 2025): what its public CT data and code contain, what our own exploration of the 60 scans shows, and what it would take to train the models ourselves.

60
planning CT scans in the public release

44
structures: 12 jaw sub-volumes + 32 teeth

0.85
best reported Dice (mandible basal)

Muaaz, Vaishnavi, Kashish, Kamakshi · Guide: Vaishali Kulkarni · Tata Memorial Hospital, Mumbai · November 2026 · github.com/MuaazSM/orn-pipeline

One mandible contour becomes 44 structures

IMPACT auto-segments the jaw on radiotherapy planning CT. Instead of treating the mandible as one organ, it splits the mandible and the maxilla into left, central and right alveolar and basal regions, and labels every tooth.

Why split the bone

- Alveolar bone around the teeth is trabecular and more radiosensitive; basal bone is denser cortical bone.
- The regions follow the ClinRad ORN staging system, which grades the vertical extent of bone damage.
- Dose can then be read per region and per tooth, for example before a dental extraction.

Task	Structures	Model in the paper
Mandible	6 sub-volumes	Swin UNETR, one stage
Maxilla	6 sub-volumes	ResUNet ROI, then Swin UNETR
Teeth	32 teeth (ADA 1–32)	ResUNet ROI, then Swin UNETR

Scope check

The scans are CT, not MRI. The release supports segmentation only: it has no clinical variables and no ORN outcomes, so it cannot be joined to our ORN-grade model directly.

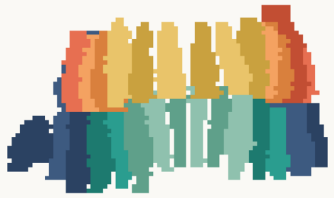
The release: 60 CT scans, each with three label maps



Mandible: 6 sub-volumes



Maxilla: 6 sub-volumes



Teeth: 32 labels

Right basal

Central basal

Left basal

Right alveolar

Central alveolar

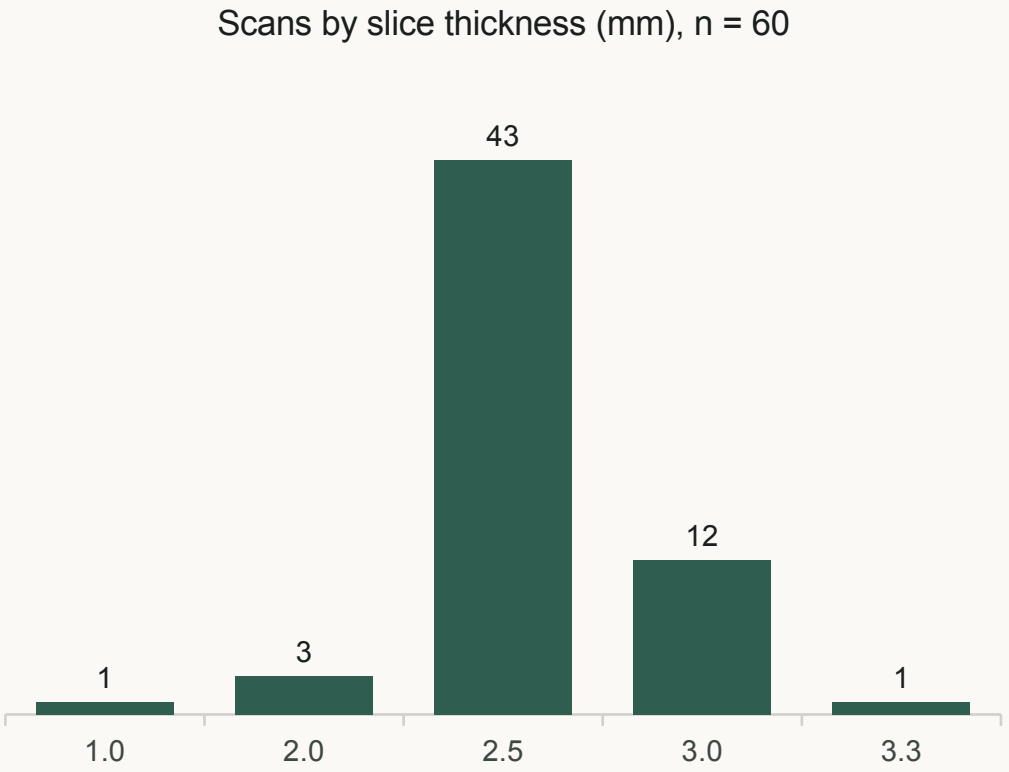
Left alveolar

Viewed from the front: the patient's right is on the left of each image. Alveolar regions sit next to the teeth, basal regions below (mandible) or above (maxilla).

File	Contents	Size
mandible.zip	60 CT + mandible masks	1.87 GB
maxilla.zip	same 60 CT + maxilla masks	1.87 GB
teeth.zip	same 60 CT + teeth masks	1.87 GB
dose_test_cases.zip	RT dose, 8 test patients	4 MB

- Every zip repeats the same CT images, so the unique imaging is 1.9 GB.
- All 180 masks sit on their CT's voxel grid (checked).
- No train/test split file, no clinical table and no ORN outcome.

Same in-plane grid, varied slice spacing and length



0.98 mm

in-plane pixels in 38 of 60 scans; the rest 1.04–1.37 mm, so resampling to 1 mm is needed

412 mm

median scan length (278–570 mm), 101 to 296 slices

25 M

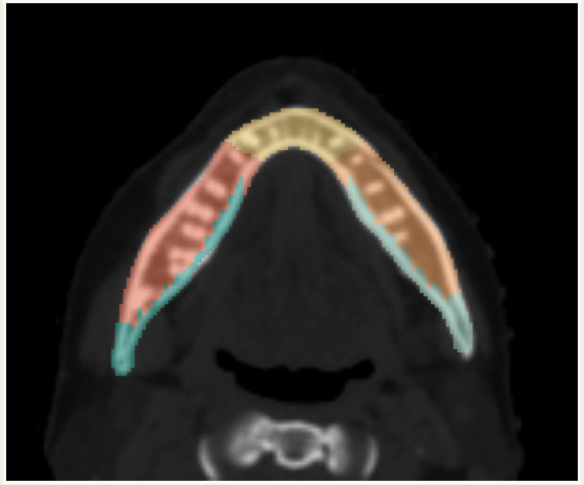
voxels per scan after the paper's resampling and body crop (14–36 M)

40 of 60

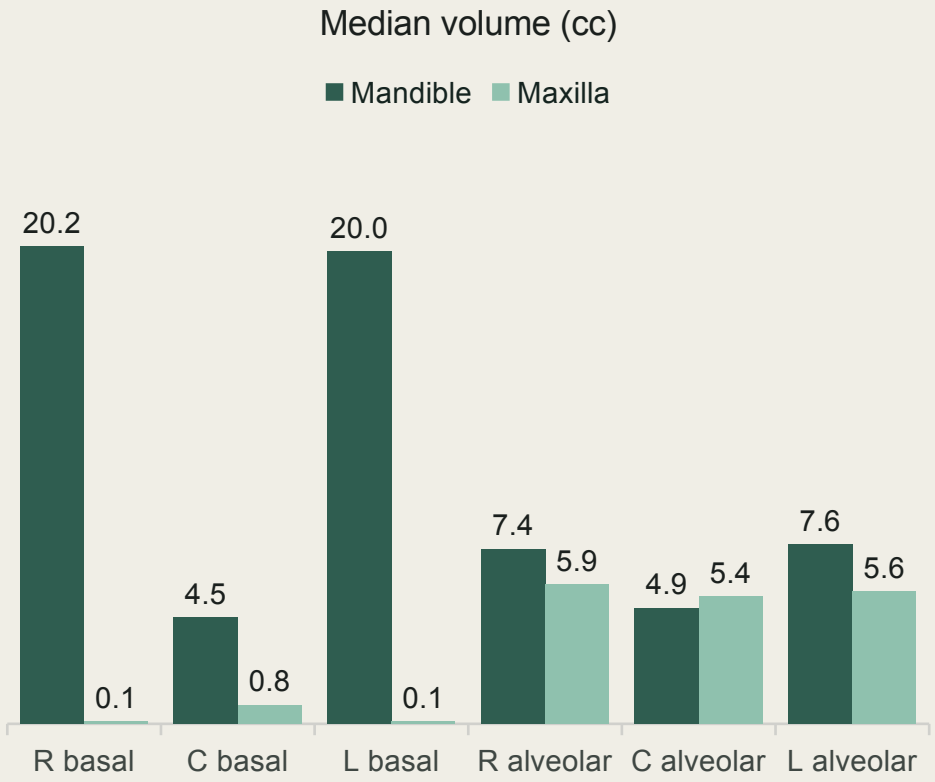
scans have metal-range voxels (> 3,000 HU) in the jaw, so streak artefacts are in the training data

All 60 scans read voxel by voxel · resampling and body crop follow the authors' code (1 × 1 × 2 mm, foreground HU > 0)

Maxilla basal bone is missing from half the scans



Axial CT with the six mandible sub-volumes overlaid



Region	Mandible	Maxilla
R basal	60	**26**
C basal	60	**29**
L basal	60	**25**
R alveolar	60	60
C alveolar	60	60
L alveolar	60	59

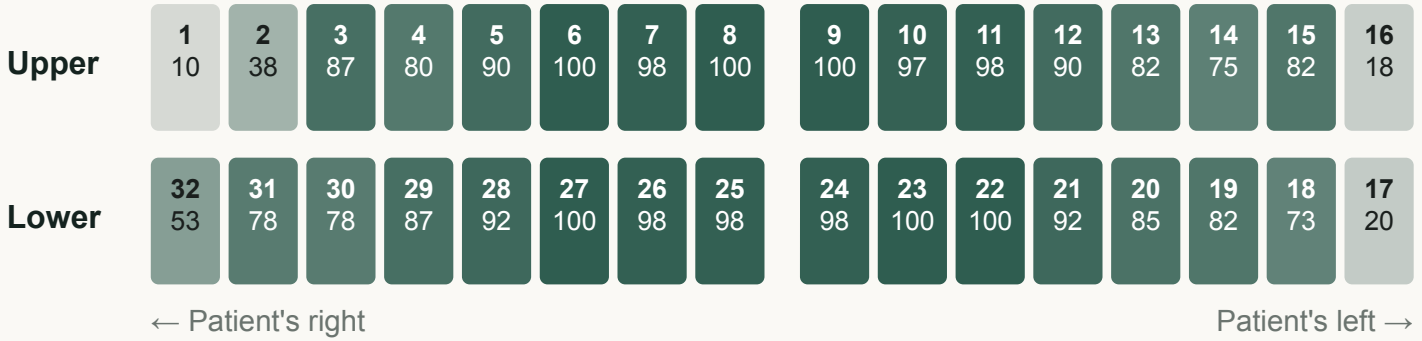
Scans (of 60) containing each region

Lateral maxilla basal regions are 0.1 cc at the median. The paper reports that these were not segmented successfully.

Median volume over scans where the region is present · side from the NIfTI orientation · axial slice of patient 3

42% of molar positions are empty

% of patients with each tooth present (ADA Universal numbering 1–32)



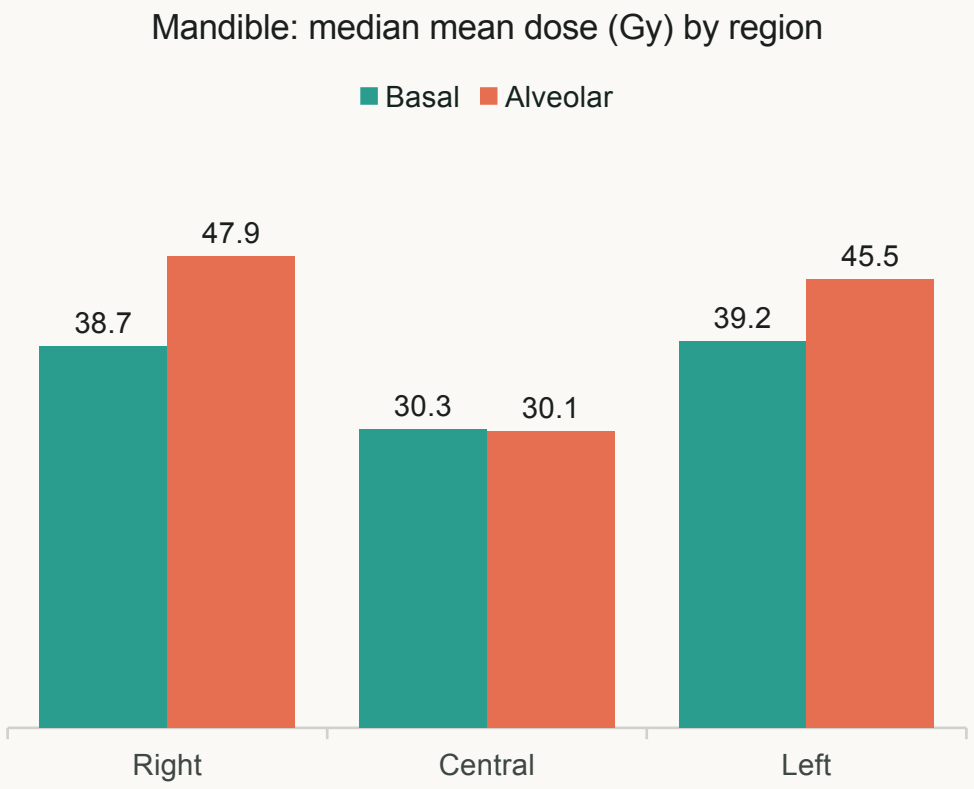
Median 27 of 32 teeth per patient (range 14–32); 6.2 missing on average, matching the paper's Table 1. Third molars (1, 16, 17, 32) are present in only 25% of patients.

Group	Present	Dice
Upper central	99%	0.80
Upper premolars	85%	0.69
Upper molars	**52%**	0.76
Lower central	99%	0.76
Lower premolars	89%	0.70
Lower molars	**64%**	**0.63**

Lower molars are both the least often present lower teeth and the lowest Dice. Upper molars do better than their presence suggests, so missing teeth are one factor, not the whole story.

Present = tooth label non-empty in the mask (n = 60) · Dice from the paper's Table 2 (test set, n = 10)

Jaw dose is lopsided: up to 42 Gy left to right



Case	Right basal (Gy)	Left basal (Gy)	Gap (Gy)
15	**14.4**	**56.0**	**41.6**
34	**47.7**	**7.7**	**40.0**
40	46.8	26.3	20.5
36	34.3	44.1	9.8
38	44.1	53.5	9.5
35	21.4	22.7	1.3
37	38.7	39.2	0.5

A single whole-mandible mean would hide this. Splitting by side, and alveolar from basal bone, keeps it.

Data issue

Case 32's dose file peaks at 16,384, about twice the prescription, with a quarter of voxels above 80 Gy. Likely a scaling error, so it is left out.

Mean dose (Gy) of the plan resampled onto the manual masks · 7 of 8 dose cases (case 32 excluded, see text)

Three models share one preprocessing chain

01 — Convert
CT and RT structures from DICOM to NIfTI; overlaps go to the central region

02 — Resample
1 × 1 × 2 mm voxels, then crop to the body outline

03 — Normalise
Clip HU to −200…1,600 to show bone, rescale to 0–1

04 — Augment
Rotate ±12°, scale ±10%, intensity ±10%; four random 96³ patches per scan

Mandible	**Swin UNETR** whole scan, 7 labels			→ **6 sub-volumes**
Maxilla	**Stage 1 · ResUNet** maxilla as one ROI	→ **Crop 128³** at ROI centre of mass	→ **Stage 2 · Swin UNETR** 7 labels	→ **6 sub-volumes**
Teeth	**Stage 1 · ResUNet** all teeth as one ROI	→ **Crop 128³** at ROI centre of mass	→ **Stage 2 · Swin UNETR** 33 labels	→ **32 teeth**

TRAINING 5-fold CV on 50 scans · 500 epochs · Dice + cross-entropy loss, lr 10^{-4} → majority vote of the 5 fold models → 10 test scans: Dice, mIoU, HD95, ASD; dose vs manual (Wilcoxon, Bonferroni)

As described in the paper, sections 2.3–2.6 · MONAI (PyTorch) throughout

The public code is a snapshot, not a pipeline

Topic	Paper says	Public code does
Network	Swin UNETR	SwinUNETR import is commented out; builds MONAI UNETR or a residual UNet
Epochs	500	500 for mandible and maxilla; 1,000 in both teeth scripts
Stage-2 crop	From the stage-1 output	Training and validation crops are centred on the ground-truth mask
Test-time steps	Majority vote; Dice, mIoU, HD95, ASD; dose	None of these are in the repository
Data split	Random 50 / 10	Depends on unsorted os.listdir order (seed 0)
Paths and hardware	—	Hard-coded MD Anderson cluster and Windows paths; CUDA or CPU only
Packaging	—	No README, requirements, trained weights, inference script or resume

What this means for us: the repo shows the authors' settings, but we have to write the pipeline that joins the stages, runs the ensemble and computes the metrics.

Audit of all five seg_model.py files in github.com/dr-mnaser/orodental-segmentation (1 commit, June 2025)

Training IMPACT takes weeks on a laptop, not a day

25
training runs: 5 stage models × 5 folds

17,500
epochs in the code (12,500 per the paper)

5.8 M
96^3 patch passes: 2.8 M training + 3.0 M for validation

15 ms
per patch, the most we could spend to finish in 24 hours

Hardware	Seconds / patch	Time
Laptop CPU (how the code runs on a Mac)	1–3	**67–200 days**
Apple M-series GPU (needs code changes)	0.1–0.3	**7–20 days**
One data-centre GPU, e.g. A100	0.01–0.03	16–48 hours
GPU cluster, folds in parallel (authors)	—	hours per model

Validation: 9.0 M sliding windows, counted at a third of a training pass (forward only).

Other blockers

- The scripts cache 50 resampled scans in memory: about 10 GB, on an 18 GB laptop shared with the OS.
- No trained weights are published, so we cannot skip training.
- There is no resume, so a laptop going to sleep restarts the run.

Workload counted from the authors' scripts and our measured scan sizes · seconds per patch are planning estimates, not benchmarks

Five steps from this study to a working model

STEP 1 · DONE

Data and audit

Release downloaded, all 60 scans explored, code audited (this deck)

STEP 2

Clean pipeline

One configurable MONAI codebase: saved split, Swin UNETR as in the paper, CUDA / Apple GPU / CPU, resume

STEP 3

Mandible first

One-stage model on a cloud GPU: one fold at reduced epochs to check, then all five

STEP 4

Two-stage models

Maxilla and teeth, cropping stage 2 from stage-1 predictions at test time

STEP 5

Evaluate

Dice, mIoU, HD95, ASD against Table 2; dose on the 7 usable dose cases

Proposed code layout

```
impact_study/
  configs/        one YAML per task and stage
  src/            data, transforms, models, train,
                  infer, evaluate, dosimetry
  outputs/        splits, checkpoints, metrics
```

How it connects to our ORN work

- Region-level dose (mean and max per sub-volume and tooth) can become new features for the ORN-grade model.
- That needs a cohort with both CT and ORN outcomes, for example TMH data; the IMPACT release has no outcomes.
- Free cloud GPUs end sessions after hours, so every run must save checkpoints and resume.

Step 1 is complete; steps 2–5 need GPU access

What a successful reproduction should match

Mean Dice to reproduce (paper, Table 2)

Mandible basal	**0.85**
Mandible alveolar	**0.82**
Maxilla alveolar	**0.78**
Upper central teeth	**0.80**
Upper premolars	**0.69**
Upper molars	**0.76**
Lower central teeth	**0.76**
Lower premolars	**0.70**
Lower molars	**0.63**

Lower molars: the abstract says 0.71, Table 2 says 0.63.

Risks

- Only 10 test scans, so every Dice has wide uncertainty.
- Maxilla basal bone and molars are rare in the data.
- Paper and code disagree on network and epochs; we will ask the authors.

Next steps

- Get GPU access (college server, TMH or cloud credits).
- Build the clean pipeline and train the mandible model first.
- Read the MRI segmentation paper our guide will share, and compare.